Troubleshooting Python Code

Northwestern

Errors are normal.
Debugging is normal.

The takeaway of this workshop:

Identify the problem before searching for the solution.

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- It will save you time over the course of a project
- Introducing solutions before you understand the underlying problem can introduce new problems
- You will actually learn!

Use logic to identify the problem.
Follow your instincts,
but don't blindly trust them.

Use logic to identify the problem.

- Don’t start trying solutions to what you think is the problem without confirming it
- Start narrowing down where the problem is

What is instinct?

What is instinct?

- “an innate, typically fixed pattern of behavior in animals in response to certain stimuli.”

What is instinct?

- Our "instinct" can be led by our insecurities.
- I'm bad at X, so X must be causing the problem.

What is instinct?

- The more Python code I see, the more types of data, the more types of errors, the better my “instinct” becomes.

First division in troubleshooting

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It used to work, but now it doesn't.

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These are two very different things.

First division in troubleshooting

I just wrote it, and it doesn't work.

It used to work, but now it doesn't.

I just made changes, and now it doesn't work

I just wrote it, and it doesn't work.

I'm getting an error message → syntactic error

It's not doing what I want it to → semantic error

I’m getting an error message → syntactic error

- Python 3.10+ has better error messages than previous versions!
- Look up the error message online or ask an LLM. Does anything seem right?
- Start debugging (we’ll talk about what this means).

It’s not doing what I want it to → semantic error

It’s a code problem

It’s a data problem

It’s not doing what I want it to → semantic error

It’s a code problem

It’s a data problem

Both will probably require debugging, but you might do it a bit differently if you have data.

It’s a data problem

1. Think about it. Could differences in your data cause the problem? (rows not formatted the same, missing data, distributions vary, etc.)
2. If you’ve identified a possible data involvement:
   - Look at the raw data (briefly), or
   - Write Python code to search through the data for the anticipated problem.
3. Try creating a shorter dataset with only a few clean rows – if that works but the full dataset fails, the problem lies with your data.

It’s a code problem

Debugging – find *where* the problem is

1. Work in a copy of your script or notebook
2. Start at the top of the script or notebook (or code cell if you know where trouble begins)
3. Write print statements and/or run the code in chunks until you get the error
   - If necessary, you can comment out code chunks, and run the code again to see if that helps identify the location of the error.
4. Check that each code chunk does what you expect (e.g., using print statements), working top to bottom, ruling out chunks that perform properly.

It used to work, but now it doesn't.

- Did you change anything in the code recently?
- Are you using a different data set?

- Are you running it on a different system (local vs. server, script vs. Jupyter notebook, local Jupyter vs. Colab, new laptop, etc.)?
- Has it been a while since the last time you ran it?

It used to work, but now it doesn't.

- **Did you change anything in the code recently?**
  - *Start debugging just above the code you changed.*

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  - *How is the new dataset different from the old?*

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  - *Can you test it again on the old system to confirm?*
  - *Rewrite the code OR create a conda environment to replicate the old system*
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  - *Python updated? Package updated? (might be a dependency)*
  - *Rewrite the code OR create a conda environment to replicate the old version*

Once you’ve identified where the problem is

1. Look it up online or ask an LLM – are there solutions?
2. Spend time thinking about how to fix it
3. Work in a copy of your script or notebook
4. Try one solution at a time and revert to original before trying each new solution
5. Ask for help if you can’t figure it out within a reasonable time

If you have a data problem:

Should you clean the data OR change the code to accommodate the messiness?

- Do you plan on running the same code on a different dataset in the future, or is this a one-time thing?
- Which will take less time?

Questions?